



A robust embedded biometric authentication system based on fingerprint and chaotic encryption



M.A. Murillo-Escobar^a, C. Cruz-Hernández^{b,*}, F. Abundiz-Pérez^a, R.M. López-Gutiérrez^a

^a Engineering, Architecture and Design Faculty, Autonomous University of Baja California (UABC), Ensenada, BC, Mexico

^b Electronics and Telecommunication Department, Scientific Research and Advanced Studies Center of Ensenada (CICESE), Ensenada, BC, Mexico

ARTICLE INFO

Article history:

Available online 28 June 2015

Keywords:

Chaotic encryption
Embedded expert system
Fingerprint template protection
Microcontroller
Security

ABSTRACT

Fingerprint recognition is a reliable solution in user authentication systems. Nevertheless, the security and secrecy of the users data are a concern in today's biometric systems and most of the security attention is focused in biometric template protection to avoid identity theft. In recent years, several approaches have been presented where the main objective is to have the biometric revocable, but almost all them have failed in verify their security and robustness. In last years, chaotic systems have been proposed in cryptography due they have several properties related with cryptography properties such as extreme sensibility on initial conditions with confusion and ergodicity with diffusion. In this paper, we present a novel fingerprint template protection scheme based on chaotic encryption by using the logistic map and Murillo-Escobar's algorithm (Murillo-Escobar et al., 2014). In addition, we present a novel implementation of our scheme in a 32 bit microcontroller for secure authentication systems to show its application on embedded expert systems. In contrast with recent approaches presented in literature, we present a complete security analysis in both statistical and implementation level, to justify the proposed scheme in a real application. Based in the results, the proposed embedded authentication system is secure, effective and at low cost, and it could be implemented on real secure access control systems.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

A biometric identifier is a physiological or behavioral characteristic of a person such as face features, iris, palm, fingerprint, hand or finger geometry, retina, voice, handwritten signature, keystroke dynamics, etc. Basically, all them are unique, unalterable, universal and can be measured. Due these properties, biometrics have been used to identify or authenticate users in a biometric system, which is becoming more popular than traditional identification techniques such as identification card (ID), password and personal identification number (PIN) (Jain & Uludag, 2003; Soutar, Roberge, Stoianov, Gilroy, & Vijaya Kumar, 1999).

Biometric systems are based on the following steps: first, a biometric sample is taken from an individual, for example iris scan; then, this biometric can be presented as an image or some data are extracted from that sample which constitute a biometric template; finally, the biometric data, either the image or the template or both, are stored on a database (local or remotely) or in a smart

card. All these steps, constitute the process known as enrollment. One time a person performs the enrollment process, he/she can identify (distinguish an individual from a larger set of individual biometric records, i.e. one-to-many matching) or authenticate (a live biometric sample presented by a person is compared with a stored sample, i.e one-to-one matching) by presenting his/her biometric sample in the system, which will compare the submitted sample with stored sample and if the matching process succeeds, the person is recognized and the system will *accept* him/her. In other case, she/he is not recognized, i.e. will be *rejected*. Talking about the reliability of a biometric system, the submitted sample (actual) can present some variance with the stored reference sample, which could make of the comparison, matching and identification, an inexact process, i.e. the biometric system does not have 100% of accuracy. There are two metrics to measure the accuracy of a biometric system, *false rejection rate* (FRR) and *false acceptance rate* (FAR). FRR is how many rejects (incorrectly) performs the biometric system, i.e. when the system rejects a legitimate user. FAR is how many acceptances (incorrectly) performs the biometric system, i.e. when the system incorrectly match a biometric sample with a wrong stored reference sample, resulting in misidentification. The ranges of FRR for most biometric system are from 0.1% to 20%, i.e. a legitimate user will be rejected one of 1000 times

* Corresponding author.

E-mail addresses: murillo.miguel@uabc.edu.mx (M.A. Murillo-Escobar), ccruz@cicese.mx (C. Cruz-Hernández), abundizf@uabc.edu.mx (F. Abundiz-Pérez), roslopez@uabc.edu.mx (R.M. López-Gutiérrez).

and one to 5 times on average. About FAR ranges are from one in 100 to one in 10,000,000 for very high security applications (Cavoukian & Stoianov, 2007).

Nevertheless, the biometric systems present some security concerns related with secrecy of personal data and identity theft. According with Ratha, Connell, and Bolle (2001) and Jain, Nandakumar, and Nagar (2008), basically there are four types of attack in a generic biometric system: attack on sensor, attack between modules, software attack and attack against the biometric template stored in database. The last kind of attack is the most dangerous in a biometric system because the biometric template can be replaced by an impostor to compromise the security of the system. In addition, the user will lost its biometric trait for ever (identity theft) and he/she cannot use it further for authentication due it is irrevocable, i.e. the authorized user can not generate two different fingerprint trait from the same finger. Therefore, biometric template protection is required in a biometric systems to increase the security in both the biometric system and personal biometric protection (Admek, Matsek, & Neumann, 2015; Kevenaar et al., 2010).

Due to these security challenges, there are an increasing interest to design, built and implement secure biometric template protection in biometric systems. *Feature transformation* and *biometric cryptosystems* methods have been proposed in literature for biometric template protection with some advantages and limitations (Jin, Beng Jin Teoh, Song Ong, & Tee, 2012), where the main objective is to keep revocability, i.e. if the biometric template is compromised, the authorized user can create another one from the same biometric trait for future use. Nevertheless, the security evaluation of such schemes is often conducted in various inconsistent manner (Inuma & Otsuka, 2013). There are four major requirements when a biometric template protection algorithm is designed (Jain et al., 2008):

- **Revocability.** Ability of the algorithm to cancel the compromised template and generate a new one from the same biometric trait. It is known as cancellable.
- **Diversity.** The secure biometric template must not allow cross-matching across databases, thereby ensuring the users privacy.
- **Security.** Extract the original template from the protected template must be computationally difficult to achieve. This ensure the secrecy of the user's biometric template.
- **Performance.** The protecting algorithm should not affect the recognition performance of the biometric system, i.e. the FAR and FRR rates need to be acceptable.

Feature transform approach is based on a transformation function F (algorithm), which is applied to the biometric template T and the result is stored in the database. The template transformed is characterized by a secret key K or a password. The matching process is performed in the *transformed domain*. Techniques such as *salting* and *non-invertible transforms* have been proposed in literature, such as robust hashing and cancellable biometric (Jain, Nandakumar, & Nagar, 2012; Munaga & Santhosh Kumar, 2014; Ratha, Chikkerur, Connell, & Bolle, 2007; Teoh, Goh, & Ngo, 2006). The security of *salting* schemes is based on the secrecy of the key or password. Whereas, the security of *non-invertible transformation* techniques uses one way functions (computationally difficult to invert), even if the key is known. Nevertheless, the major disadvantage of these techniques is that security of the system is difficult to verify due there is not mathematical foundation to perform a robust security analysis and it is assumed that the distribution of biometric features is uniform (Kaur, Sofat, & Saraswat, 2010) and an adversary may be able to exploit the non-uniform nature of the biometric features to launch an attack that may

require significantly fewer attempts to compromise the system security (Jain et al., 2008).

On the other hand, *biometric cryptosystems* schemes protect a cryptographic key with the biometric or generate a cryptographic key from the biometric by using fuzzy vault or secure sketch techniques (Cavoukian & Stoianov, 2007; Teoh & Kim, 2007; Uludag, Pankanti, Prabhakar, & Jain, 2004). Some public information is stored in database called *helper data*, which is required during matching process to extract the cryptographic key from the query biometric features. The original biometric template never is saved on database, just the cryptographic key which is used in matching process. Techniques such as *key binding* and *key generation* have been proposed in literature (Imamverdiyev, Beng Jin Teoh, & Kim, 2013; Moujahdi, Bebis, Ghouzali, & Rziza, 2014). Virtually, all types of biometrics have been tested to generate a digital key, i.e. fingerprint, iris, face, keystroke dynamics, voice, handwritten signatures, palmprints, and acoustic ear. Of all these biometrics, the most promising result have been achieved with iris with $FFR = 0.47\%$ and $FAR = 0$ to generate a 140 bit key. These error are marginally near of conventional iris biometric system. Biometric cryptosystems based on fingerprint present a noticeable degradation in accuracy compared with conventional fingerprint biometric systems. In resume, the security of these schemes is based on helper data, which is computationally difficult to recover both the cryptographic key or the original template. Nevertheless, key generation biometric cryptosystems with high key entropy and stability have been more difficult to implement in practice (Abhyankar, Vijayat, Kumar, & Schuckers, 2009; Arakala, Jeffers, & Horadam, 2007; Chang & Roy, 2007; Nandakumar, Nagar, & Jain, 2007; Sutcu, Li, & Memon, 2007).

Steganography, watermarking and cryptography are other approaches to protect sensitive information against not authorized users. Steganography is a technique where the secret message is hidden or embedded within another unrelated message, for example hide text in a host image. Whereas, watermarking is used for the copyright protection (Khan, Zhang, & Tian, 2004). Cryptography is a technique thoroughly used on secure communications and private data storing, where the plain data are transformed in encrypted data (unrecognisable for not authorized users) by using an algorithm and secret key. Although, these schemes do not guarantee revocation and they are not recommend for biometric protection because the matching process requires decryption of stored templates and the security depends of the cryptographic key (Al-Assam, Sellaheewa, & Jassim, 2009; Jain et al., 2008), the cryptographic method can achieve a very high security and it can be verified with several security analysis at statistical level (where several approaches fail).

Currently, conventional cryptography such as Triple Data Encryption Algorithm (TDEA) and Advanced Encryption Standard (AES) are used as symmetric ciphers (one secret key to encryption and decryption) where plain data are encrypted in blocks. TDEA and AES are based in a Feistel network and a permutation–diffusion network, respectively. Nevertheless, TDEA is slower than AES to encrypt data and AES has drawback in the number of permutation–diffusion rounds. In addition, AES has an encryption algorithm highly structured that could be broken in the next years according with Ferguson, Schroepel, and Whiting (2001). Rivest, Shamir and Adleman (RSA) is an asymmetric encryption algorithm (it uses two secret keys for encryption and decryption) with security advantages but it is slow to encrypt data than any other symmetric algorithm. Recently, DNA cryptography has been proposed (Abbasy, Manaf, & Shahidan, 2011; Adleman, 1994; Bordoia & Tornea, 2010; Guozhen, Mingxin, Lei, & Xuejia, 2006; Taylor, Risca, & Bancroft, 1999; Xuejia, MingXin, Lei, JunSong, & XiWen, 2010), but these methods are not effective due the influence of the

biological factors (Zhou, Zhang, Wei, & Zhou, 2010). In contrast, chaotic cryptography has received much attention in recent years due the chaotic systems present interesting properties such as extreme sensitivity to initial condition and control parameter, mixing data, ergodicity, kind of pseudorandom behavior, determinism, etc. According with Shannon (1949), these properties are highly related with cryptographic properties to design excellent encryption algorithms with excellent confusion and diffusion process, and complexity in source system. In recent years, chaotic maps have been employed principally for image encryption at MatLab simulation level, where most of the proposed schemes use the diffusion and confusion architecture (Bhatnagar, 2012; Bhatnagar & Jonatan Wu, 2012, 2015; Cui, 2010; Han, Hu, Yu, & Wang, 2007; Hsiao & Lee, 2015; Inzunza-González & Cruz-Hernández, 2013; James & Philip, 2012; Khan, Zhang, & Tian, 2007; Liu, 2012; Murillo-Escobar, Cruz-Hernández, Abundiz-Pérez, López-Gutiérrez, & Acosta Del Campo, 2015; Zhao, Li, & Yan, 2008).

Literature presents some advances of template protection based in chaos and features transform techniques. In Supriya and Manjunatha (2014), a new approach based on cancellable biometrics and chaos is presented. The biometric sample is transformed and stored in database by using chaos derived from logistic map. The matching process is performed in transformed domain. In this scheme, the authors present a key space analysis related with initial condition and parameter control of the logistic map and an entropy analysis of the transformed biometric sample, but the security analysis presented is insignificant to verify the effectiveness and security of the proposed scheme in real application. In Mihailescu (2014), a new enrollment scheme for biometric template based on hash chaos-based cryptography is proposed. The scheme uses the logistic map and Rossler system to generate the session key used in the hash function. Nevertheless, the authors do not present a security analysis of their proposed scheme. In Ghouzali and Abdul (2013), the authors present the design and development of a new approach for biometric template protection based on the chaotic behavior of logistic map. In their approach, a protection function $P(.)$ is used to mix the plaint template with the chaotic random matrix using the user key entered by the he/she. Therefore, it is a feature transform scheme by using the logistic map. Nevertheless, several specifications and analysis are not presented such as algorithm details, key management and security analysis. In Li, Wang, Zhang, and Wu (2012) and Li and Zhang (2010), a chaos-based cancellable palmprint authentication algorithm for palmprint template protection is presented. In both articles, the authors use chaotic stream cipher to generate a cancellable palmprint template. The scheme presents some security analysis such as key space, accuracy and speed with excellent result. Nevertheless, these analysis are not enough to show and verify the effectiveness of their proposed. In Xiaomin, TaiHua, and Wenfang (2011), the authors present a biometric template protection based on fuzzy extractor and chaos for biometric authentication. Although the authors says their scheme is robust against attacks and can be implemented in real biometric applications, just the FAR and FRR accuracy analysis was performed. Khan and Zhang (2005) proposed a novel chaotic encryption method to protect and secure biometric template, which uses two logistic maps for the encryption/decryption process and Henon map for template encryption. Although the authors remark that their scheme is secure, fast and easy to implement, they do not present any security analysis. Rigorous security analysis must be performed to those available in the cryptanalysis literature before deployed in critical real-world applications such as in criminology, tourism, counter terrorism measures, control of movement of persons, attendance and access control, data protection, electronic banking, etc. (Jain et al., 2008).

On the other hand, most of the biometric systems are implemented in embedded systems. These systems are based on software and hardware, which uses a processor with limited computational power, limited memory and input–output peripheral. The low cost, low physical size, low power consumption, high performance and flexibility are some advantages of such systems. The embedded systems we use every day are designed to be simple to use and effective in their operation, but more complex software have been required in the last years to help people to take decisions faster and effectively over a specific event. Therefore, there is an increasing interest to design and development intelligent real-time control systems or sophisticated systems (embedded expert systems) with high speed process, continuous operation without delays, focused attention, direct interface with sensors and devices, use of embedded hardware such a microcontroller with limited resources and ability to share data effectively, effective communication within processors, etc. In this sense, an embedded authentication biometric system with a template security interface can help to companies reduce the risk of fraud and identity theft. If these systems have “intelligence” capacity to perform the biometric enrollment and authentication with security guaranties, low cost, high performance, template protection guaranteed, store data and transmit it over insecure channels, then it could be considered as a basic embedded expert system.

In literature, some implementations of biometric authentication systems based on microcontroller and FPGA (Field Programmable Gate Array) are presented, but any biometric template protection is achieved. Two microcontroller implementation of biometric systems based on fingerprint and voice are presented in Chen and Dai (2005) and Nayak (2008), respectively. Although the analysis show high performance and efficiency of the author's method, the biometric data are not protected and the user's template is exposed. Another biometric authentication system implementation based on FPGA is presented in Militello, Conti, Vitabile, and Sorbello (2008) and Nie, Li, Zhou, and Toyonaga (2013), respectively, where fingerprint trait is used. Nevertheless, biometric data encryption are not achieved in any FPGA implementation. Another implementation where microcontroller-FPGA co-design are used for fingerprint authentication system are presented in Fons, Fons, and Cantó (2007) and Wang, Li, Isshiki, and Kunieda (2004), but user identity is exposed.

In this paper, we proposed a novel embedded fingerprint authentication system implemented in a 32-bit microcontroller with biometric template protection by using an encryption algorithm based on chaos. The proposed biometric scheme protections is based on features transform, but the matching process is performing in plain domain (decryption template is required). The implementation is based in an embedded expert system with high accuracy, secure enrollment and authentication process, with fingerprint template protection. The security of the proposed scheme is verified with a complete statistical security analysis (that most of the biometric template protection schemes in literature do not present) and a hardware analysis such as architecture, memory space, communication ports, frequency system, speed, precision, cost. All the analysis results justify the proposed implementation for critical real-world applications. The encryption algorithm is based on our recently scheme in Murillo-Escobar, Abundiz-Pérez, Cruz-Hernández, and López-Gutiérrez (2014), which is a symmetric algorithm with a 128 secret key, it uses just one permutation–diffusion round and optimized chaotic sequences from onedimensional logistic map.

The organization of this work is as follows. In Section 2, the proposed embedded system scheme, the enrollment process, and the authentication process are shown. The encryption algorithm based on previous work in Murillo-Escobar et al. (2014) is presented in

Section 3. The experimental results, security analysis and comparison with literature are shown in **Section 4**. Finally, in **Section 5** this work is concluded.

2. Authentication system

The proposed embedded system is based on the next hardware modules: 32 bit microcontroller M52259 of Freescale, fingerprint reader module Futronic FS83 and human interface (**Fig. 1**). The enrollment process can register a new authorized user in the embedded system with a fingerprint (example the right index finger). First, the fingerprint module uses minutia extraction technique to generate the user's template (2072 bytes from three samples) and send it to the microcontroller via RS-232 communication (data transmission between processors); then, the template data are encrypted by using the proposed encryption algorithm based on chaos and a 128 bit secret key (characterized by 32 hexadecimal digits); finally, the encrypted template is stored on Flash memory's microcontroller (**Fig. 2**). Once the authorized user was successfully registered in the system, further the user can access to the restricted area (fiscally or logically) by the authentication process. First, the authorized user request his encrypted template; then, the decryption process is used to recover the user's template by using the same secret key; after that, the microcontroller send the decrypted template to the fingerprint module via RS-232 communication; finally, the authentication process is achieved on fingerprint module with its matching algorithm. If the templates are the same, the user has access to the restricted area (**Fig. 3**).

The scanner of the fingerprint module is based on infrared LED technology and it has a processor with a fingerprint recognition algorithm based on minutia extraction with a *false acceptance rate* FAR 10^6 (one in a million) and *false rejection rate* FRR 10^2 (one in one hundred). The microcontroller achieve several task such as read the fingerprint template, cipher and decipher the fingerprint template, store the encrypted template on Flash memory (or transmit it over an insecure channel), request the matching process for authentication and perform the human interface (keypad, display, LED and control output signals for secure access control). The human interface helps people to enrollment and authentication process.

3. Encryption algorithm

The proposed encryption algorithm is based in our previous work (Murillo-Escobar et al., 2014, 2015). We assume a plain template $P \in [0, 255]$ of length $\ell = 2072$ (bytes) based on data read by the fingerprint module. **Fig. 4** shows the block diagram of the

encryption process. In order to transform the plain template to encrypted template, we use the pseudorandom sequences generated by a chaotic system. Several chaotic properties are strongly related with cryptography properties, which are shown in **Table 1**.

The unidimensional logistic map is known as the most simple non linear chaotic system that achieve clearly the chaos route (May, 1976). It is described as follows

$$x_{i+1} = ax_i(1 - x_i), \quad (1)$$

where $x_i \in (0, 1)$ is the discrete state, with initial condition $x_0 \in (0, 1)$ and control parameter $a \in (3.999, 4)$ to generate the chaotic sequences used in this work.

We use the unidimensional logistic map due it presents several advantages such as simple structure, easy to implement in digital systems due its discrete nature, low implementation resources, low memory consumption and high speed data generation (Cristian-Iulian & Vasile-Gabriel, 2014). Nevertheless, the logistic map has some disadvantages when it is used in cryptography such as chaotic ranges discontinues, distribution not uniform, small space key and periodicity in chaotic ranges (Arroyo, Alvarez, & Fernandez, 2008). In countermeasure, we proposed solutions in encryption algorithm to avoid disadvantages while maintaining its advantages.

3.1. Secret key definition

We use 128 bits as secret key, which is characterized by 32 hexadecimal digits $K \in [0 - 9, A - F]$. The secret key is divided in four sections to generate the initial condition and control parameter of two logistic maps (**Table 2**). Any key used is considered strong due it always produces a chaotic sequence. In addition, there are some keys stronger than others and this is analyzed in randomness analysis in **Section 4**. These considerations avoid the chaotic ranges discontinues, small space key and periodicity in chaotic ranges of the logistic map.

3.2. Calculation of Z

The Z value is very important to increase the sensitivity at bit level in both plain text and secret key. This process, increase the security against differential attack such as chosen/known plain template attack. All plain template elements are summed with chaotic data from logistic map 2 to calculate Z. First, the logistic map 2 is iterated $l = 2172$ times by using a_2 and x_{20} from **Table 2**. This generates the chaotic sequence $x^{l2} = x_1^{l2}, x_2^{l2}, x_3^{l2}, \dots, x_l^{l2}$ with $x^{l2} \in (0, 1)$ and 10^{-15} decimal precision. The chaotic sequence x^{l2} is optimized by the next expression

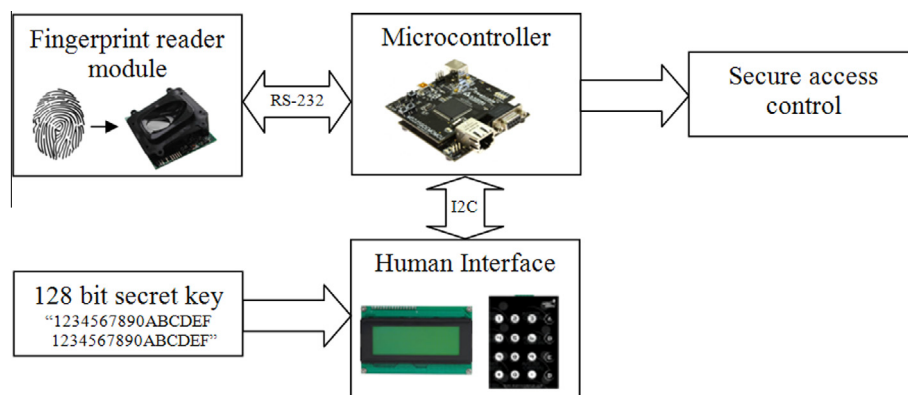


Fig. 1. Embedded authentication system scheme.

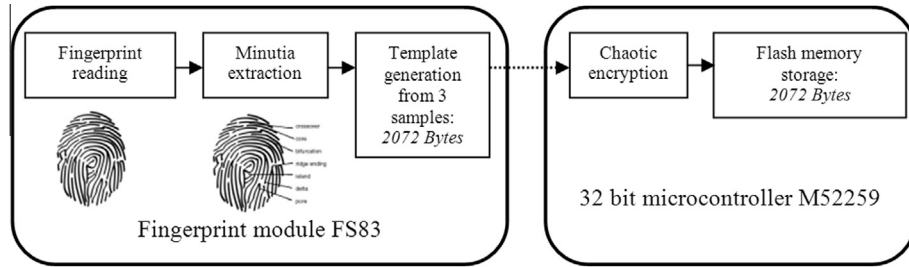


Fig. 2. Enrollment process scheme.

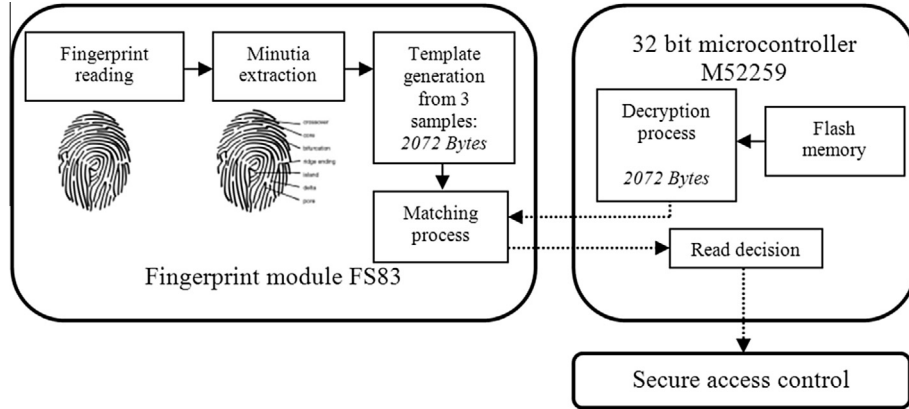


Fig. 3. Authentication process scheme.

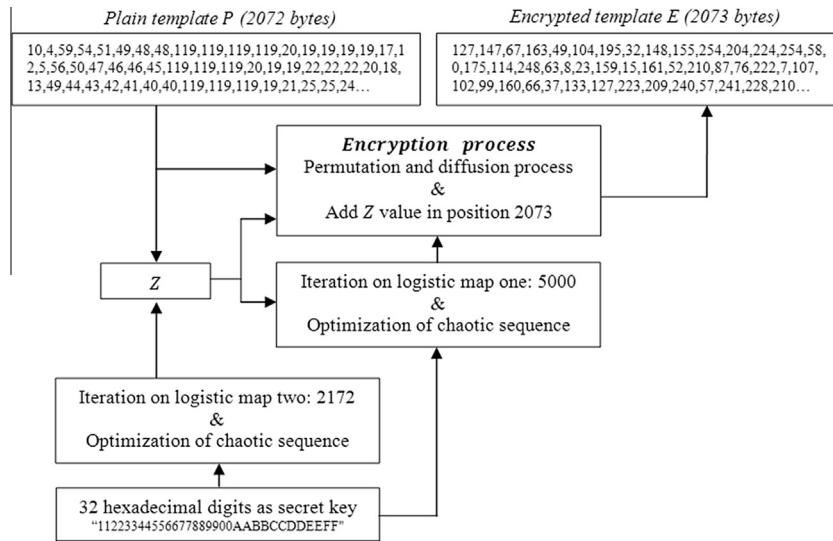


Fig. 4. Encryption scheme.

$$x_I^{L2} = \{[x_I^{L2}] * 1000\} \bmod 1, \quad (2)$$

where $I = 1, 2, 3, \dots, 2172$. Then, all plain elements are summed with x^{L2} as follows

$$S = \{S + [P_i * x_{2173-i}^{L2}] + x_{2173-i}^{L2}\} \bmod 1, \quad (3)$$

where $i = 1, 2, 3, \dots, 2072$, P_i represents the plain element i , S is a variable initialized in zero, x^{L2} is the chaotic sequence, and $\bmod$ is the module operation. Then, Z must be a value between 1 – 255 to avoid cancelation of Z , i.e. $Z = 0$. Therefore, the next operation is determined

$$V = \text{round}(S * 254) + 1, \quad (4)$$

where $V \in [1, 255]$ and round is the round to nearest operation. Finally, Z is given by

$$Z = V / 256, \quad (5)$$

where $V \in (0, 1)$ with 10^{-15} decimal precision.

3.3. Encryption process

The confusion and diffusion architecture is implemented in encryption algorithm, where all plain elements are permuted and

Table 1

Properties related between chaos and cryptography (Alvarez & Li, 2006).

Chaos property	Cryptography property	Relation between them
Ergodicity	Confusion	The chaotic data have uniform distribution and it helps to change each position of plain template elements
Extreme sensibility to initial condition and control parameters	Diffusion	A small change at input produces a big change at output, which can transform effectively the plaint template elements
Mixing data	Diffusion	A small range of initial condition can produce a lot of encrypted templates
Deterministic dynamic	Pseudorandom data	The chaotic systems produces a pseudorandom behavior, which can be reproduced if the same initial condition and control parameter are used
Complex structure	Algorithm complexity	The non linearity of chaotic systems make them difficult to resolve and it gives robustness to the encryption algorithm

transformed according with chaotic data generated by the logistic map 1. The chaotic sequence $x^{L1} = \{x_1^{L1}, x_2^{L1}, x_3^{L1}, \dots, x_T^{L1}\}$ is calculated by using $Z, T = 5,000$ iterations, control parameter a_1 , and initial condition x_{i_0} from Table 2. It has a decimal precision of 10^{-15} and $x^{L1} \in (0, 1)$. For the confusion process, a sub-sequence from x^{L1} is calculated as follows

$$Q_i = \text{round}(x_{2928+i}^{L1} * 2071) + 1, \quad (6)$$

where $i = 1, 2, 3, \dots, 2072$, $Q \in [1, 2072]$ is the pseudorandom position vector for confusion process, and *round* is the round to nearest operation. If all plain elements are permuted, then the encryption has an optimized confusion process. Nevertheless, some values are repeated by using Eq. (6). All missed position in Q are found by software and they are replaced by the repeated positions automatically from smallest to largest. For diffusion process, other sub-sequence from x^{L1} is calculated as follows

$$F_i = \{(x_{2928+i}^{L1} * 1000) + Z\} \bmod 1, \quad (7)$$

where $i = 1, 2, 3, \dots, 2072$ and $F_i \in (0, 1)$ is a vector with 2072 length with 10^{-15} decimal precision. The logistic map presents distribution not uniform due it generates many values near to 0 and 1, and it could affect the security directly Arroyo et al. (2008). In countermeasure, we amplified the x^{L1} data 1000 times with module 1 as seen in Eq. (7). Fig. 5 shows the distribution of x^{L1} and its optimization. This process increase the security of the encryption process against the powerful chosen/known plain template attack, which have broken several cryptosystems based on chaos (Alvarez & Li, 2009; Balaji & Nagaraj, 2008; Cokal & Solak, 2009; Li, Chen, &

Halang, 2009; Rhouma, Solak, & Belghith, 2010; Solak, Cokal, Yildiz, & Biyikiglu, 2010; Wang, Liao, Xiang, Wong, & Yang, 2007).

Then, F_i is transformed from $(0, 1)$ to $[0, 255]$ as follows

$$Y_i = \text{round}(M_i * 255), \quad (8)$$

where $i = 1, 2, 3, \dots, 2072$, $Y_i \in [0, 255]$ is a vector with length 2072, and *round* is the round to nearest operation. Finally, the encryption process is based on Q as pseudorandom confusion vector and Y as pseudorandom diffusion vector. The encryption process is as follows

$$E_i = (P(Q_i) + Y_i) \bmod 256, \quad (9)$$

where $i = 1, 2, 3, \dots, 2072$, P is the plain template and $E \in [0, 255]$ is the encrypted template to store in data base or transmit over an insecure channel.

The Z value must be add in the cryptogram E in the position 2073 as follows

$$E_{2073} = V. \quad (10)$$

3.4. Decryption process

First, Z is extracted from the cryptogram with the following expression

$$Z = E_{2073} / 256. \quad (11)$$

Then, x^{L1} is generated with the same secret key used in encryption process. Then, Q_i and Y_i are calculated as in encryption process. Finally, the encrypted template is decipher with the following expression

$$D(Q_i) = (E_i - Y_i) \bmod 256, \quad (12)$$

where $i = 1, 2, 3, \dots, 2072$, Q is the pseudorandom confusion vector, E is the cryptogram, Y is the pseudorandom diffusion vector, and D is the recovered fingerprint template.

4. Experimental results

In this section, a complete security analysis at statistical level is achieved. The plain template and encrypted template are extracted from microcontroller by using a USB memory stick. MatLab software is used for the corresponding security analysis such as secret key sensitivity, floating frequency, histograms, autocorrelation, information entropy, randomness, and differential analysis. On the other hand, hardware resources and performance are analyzed such as architecture, memory space, communication ports, frequency system, speed, precision and cost to verify its security and efficiency in a real system.

The proposed embedded authentication system can be considered as an embedded expert system due it has the capacity to perform the biometric enrollment and authentication with high security guaranties, at low cost and with high performance. The template protected can be stored or transmit over an insecure

Table 2

Handling of secret key (Murillo-Escobar et al., 2014).

Secret key	Control parameter	Initial condition
32 HEX digits	$H_1, H_2, \dots, H_{32}$ where $H \in [0 - 9, A - F]$	
Calculation	$A = \frac{(H_1, H_2, \dots, H_8)_{10}}{2^{32} + 1}$ $B = \frac{(H_9, H_{10}, \dots, H_{16})_{10}}{2^{32} + 1}$	$C = \frac{(H_{17}, H_{18}, \dots, H_{24})_{10}}{2^{32} + 1}$ $D = \frac{(H_{25}, H_{26}, \dots, H_{32})_{10}}{2^{32} + 1}$
Logistic 1	$a_1 = 3.999 + [(A + B + Z) \bmod 1] * 0.001$	$x_{10} = (C + D + Z) \bmod 1$
Logistic 2	$a_2 = 3.999 + [(A + B) \bmod 1] * 0.001$	$x_{20} = (C + D) \bmod 1$
Range	$3.999 < a_{1,2} < 4$	$0 < x_{10,20} < 1$
Precision	10^{-15}	
	where $(a \bmod b) = (a - b) \times (a/b)$ with $b \neq 0$	

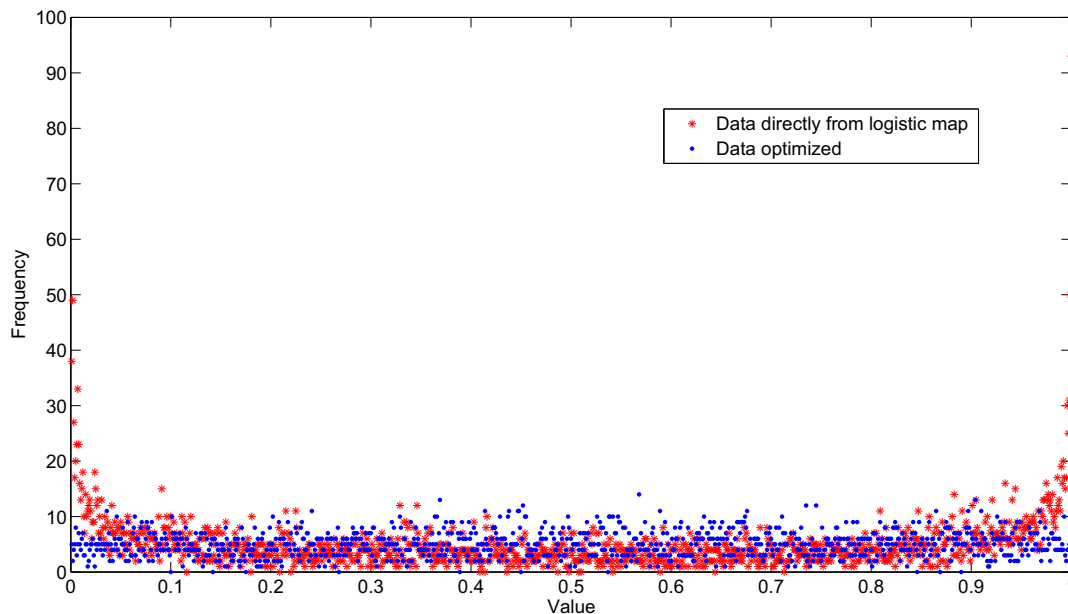


Fig. 5. Distribution data from logistic map (star) and optimized data (dot).

channel. This system could replace efficiently at people who are working in secure access control areas. In addition, the flexibility of such systems make them suitable to add real-time monitoring systems such as a digital camera with thermal sensing to ensure the users authenticity. This is proposed as future work in the conclusion section.

4.1. Key size analysis

Currently, an encryption system needs more than 2^{100} secret keys according with Alvarez and Li (2006) (considering all them strong, i.e. each key must generate chaotic data) to resist an exhaustive attack, where each possible key is tried until find the plain template. In this work, we use 128 bit secret key. Therefore, we have 2^{128} keys. The secret key handling is based in Murillo-Escobar et al. (2014), where all keys are considered strong based in maximum Lyapunov exponent analysis.

4.2. Key sensitivity analysis

A cryptographic system must be highly sensitive at small variations (at bit level) in secret key, in both encryption and decryption process (Alvarez & Li, 2006). In encryption process, when a template is encrypted by using two similar keys (just one bit different between them), the result must generate two encrypted templates highly different between them. On the other hand, just the correct secret key can recover the original template in decryption process. In this subsection, the secret key sensitivity for encryption and decryption process is tested and verified, with three similar secret keys (Table 3).

We proposed a plain template of 50 elements fixed in 150 to perform this analysis. Fig. 6 shows the secret key sensitivity at encryption process by using the secret keys of Table 3. Each

encrypted template has its own route, therefore, the encryption algorithm is highly sensitive at secret key. Fig. 7 shows the secret key sensitivity at decryption process, where just the correct secret key recover the original template (50 elements fixed in 150).

4.3. Plain template sensitivity analysis

A cryptographic system must be highly sensitive at input, i.e. small variations (one bit) of plain template (Alvarez & Li, 2006). If the encryption process generates different encrypted templates from similar plain templates (one bit different between them), even by using the same secret key, the encryption algorithm is considered highly sensitive at plain template. Two metrics called NPCR (Net Pixel Change Rate) and UACI (Unified Average Changing Intensity) are used to measure the differences between two encrypted templates E_1 and E_2 from two similar plain templates, respectively. The NPCR measures how many elements are different between E_1 and E_2 in percentage, where 100% means they are totally different; it is calculated with the following expression

$$NPCR = \frac{\sum_{i=1}^{i=2073} W(i)}{2073} \times 100, \quad (13)$$

where

$$W(i) = \begin{cases} 0 & \text{if } E_1(i) = E_2(i), \\ 1 & \text{if } E_1(i) \neq E_2(i), \end{cases} \quad (14)$$

The UACI measures how many magnitude in average is different between E_1 and E_2 , where 100% means that both of them are totally different in magnitude. It is determined by

$$UACI = \frac{100}{2073} \sum_{i=1}^{i=2073} |E_1(i) - E_2(i)|, \quad (15)$$

In this analysis, we use two plain templates with one bit of difference between them. They are encrypted to produce E_1 and E_2 by using the same secret key. The average result of eight hundred cryptograms are $NPCR = 99.5\%$ and $UACI = 88.4\%$. The result verify the high sensitivity at small changes in plain template. In addition, it has a uniform distribution according to Fig. 8. The down peak in cryptogram 285 is because there is a low probability to have the

Table 3
Secret keys used for key sensitivity analysis.

Key	Secret key
KEY 1	11223344556677889900AABCCDDEEFF
KEY 2	11223344556677889900AABCCDDEEFF
KEY 3	11223344556677889900AABCCDDEEFF

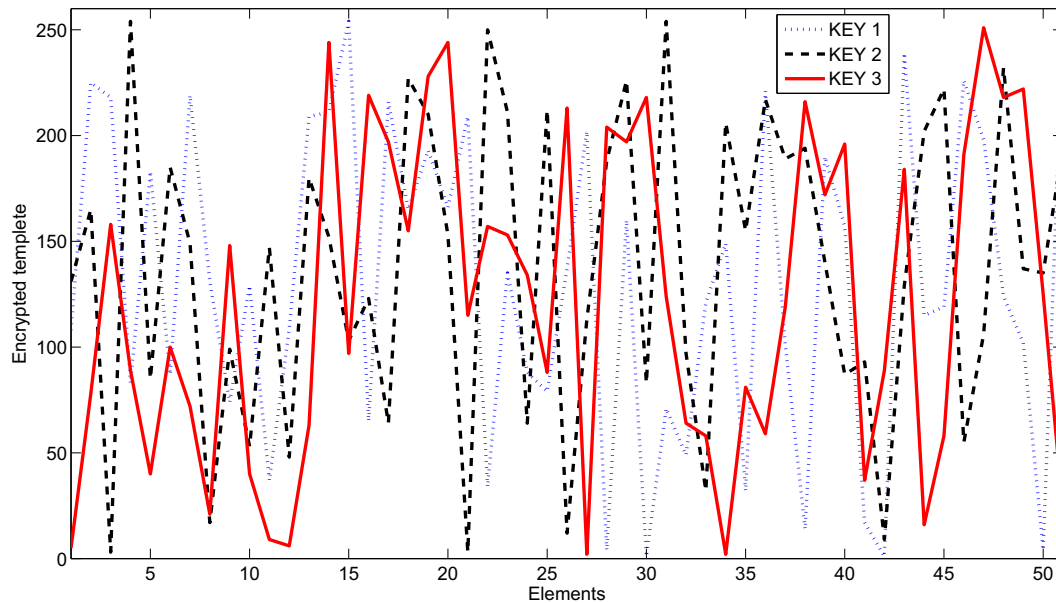


Fig. 6. Key sensitivity in encryption process.

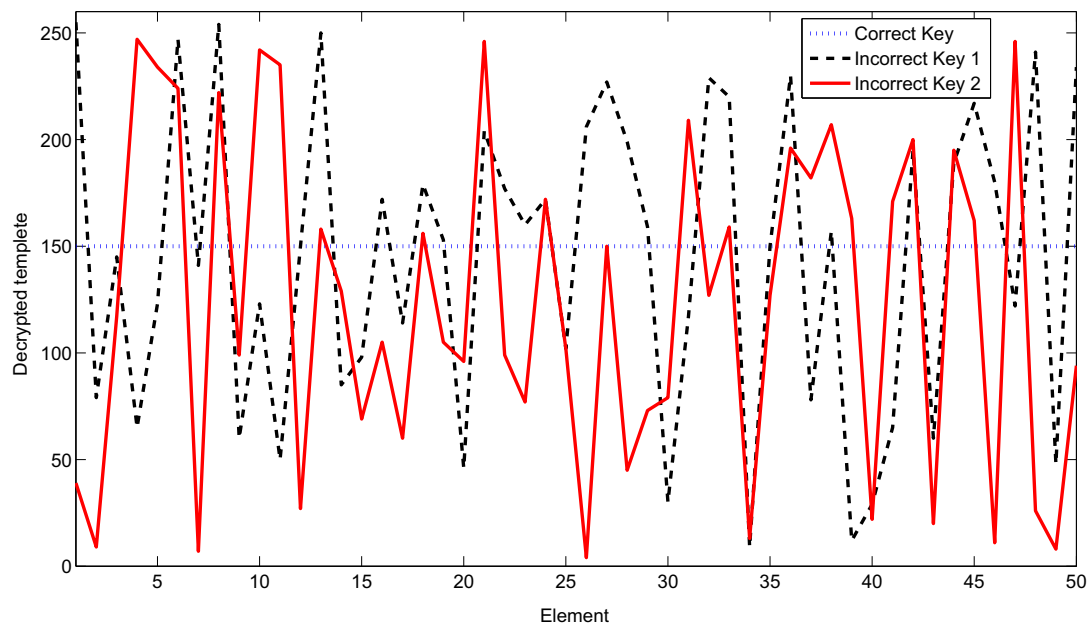


Fig. 7. Key sensitivity in decryption process.

same value of Z (even using different plain templates) but this is not a concern on the cryptographic system, because a successful decryption process does not depend just of the Z value, i.e. the correct secret key of 128 bits is required.

4.4. Floating frequency analysis

Floating frequency verifies if some section in the cryptogram is weak or in contrast, all the cryptogram has strong pseudorandom properties. If the encryption produces a weak section, it can be used to find the plain template or the secret key; therefore, the floating frequency must be uniform and it must have all possible symbols. In this analysis, we propose windows of 256 elements long; first, we count how many elements are different in each window; then, the window is shifted one position to the right and the calculation is repeated. Fig. 9 shows the floating frequency graphic

for plain template and encrypted template; it indicates that the encryption process is uniform, strong, and it does not present weak encryption sections. In a perfect encryption, each encrypted template window must have 256 different elements; the proposed encryption algorithm produces 62.5% of all possible elements against 24% of plain template.

4.5. Histogram analysis

The histogram shows the frequency of each element in a graphical form (statistical data), i.e. how much times each symbol appears in a message. The plain template has its own histogram pattern (Fig. 10(a)), but the encrypted template must have a uniform histogram to avoid any suspicious of how is the plain template (Fig. 10(b)). Therefore, the diffusion process produces a uniform histogram to resist a statistical attack.

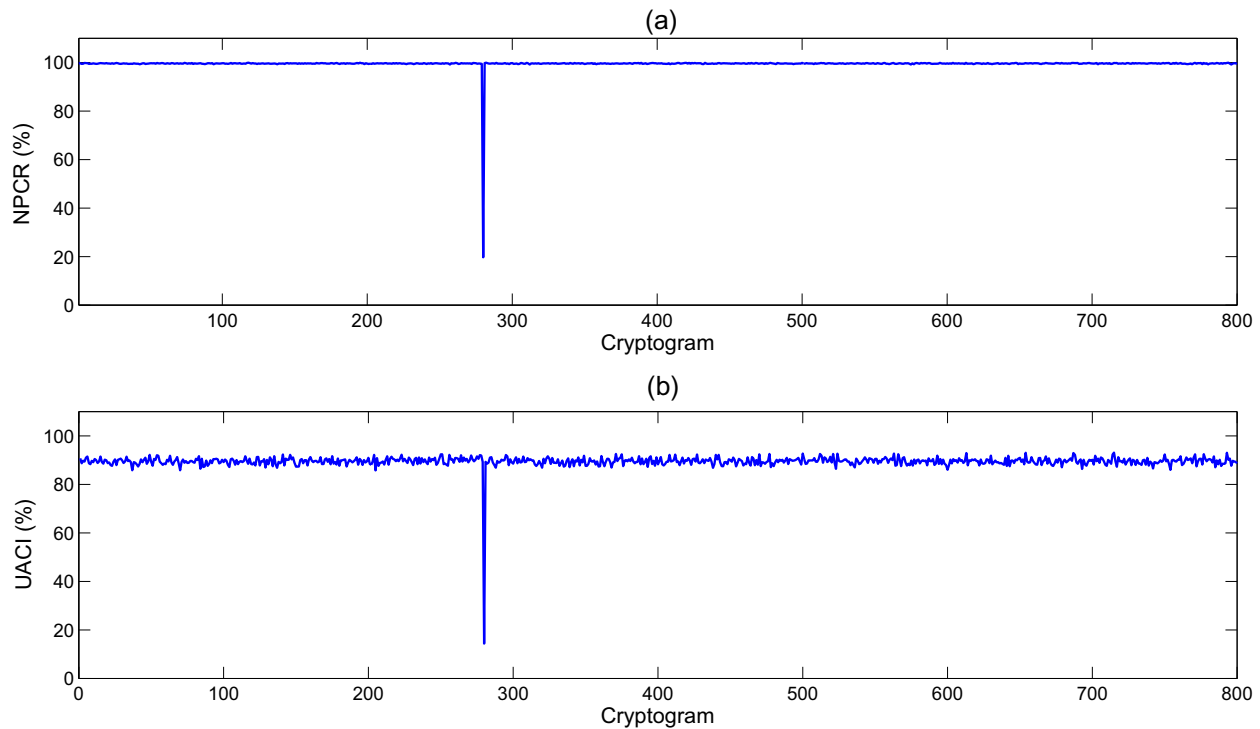


Fig. 8. Differential results from 800 cryptograms: (a) NPCR results and (b) UACI results.

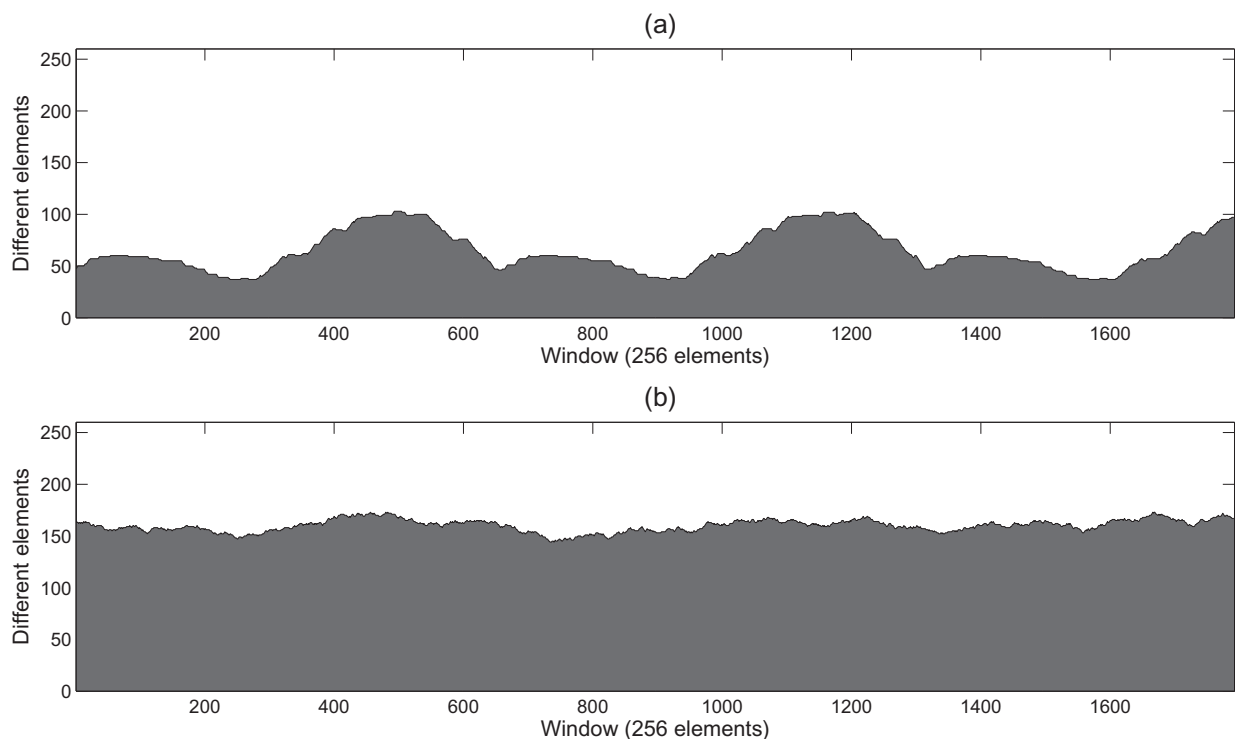


Fig. 9. Floating frequency analysis: (a) plaintext template and (b) encrypted template.

4.6. Autocorrelation analysis

The autocorrelation is defined as the correlation of a template with itself displaced t positions. With this analysis, we can determine if the encryption process generates periodic encryption or some

autodependence. Fig. 11 shows the autocorrelation graphic from Fig. 10, where the plain template (discontinuous line) shows periodic patterns, whereas the encrypted template (continuous line) does not show any periodic pattern. Therefore, the encryption process does not produce periodic encryption, it is purely pseudorandom.

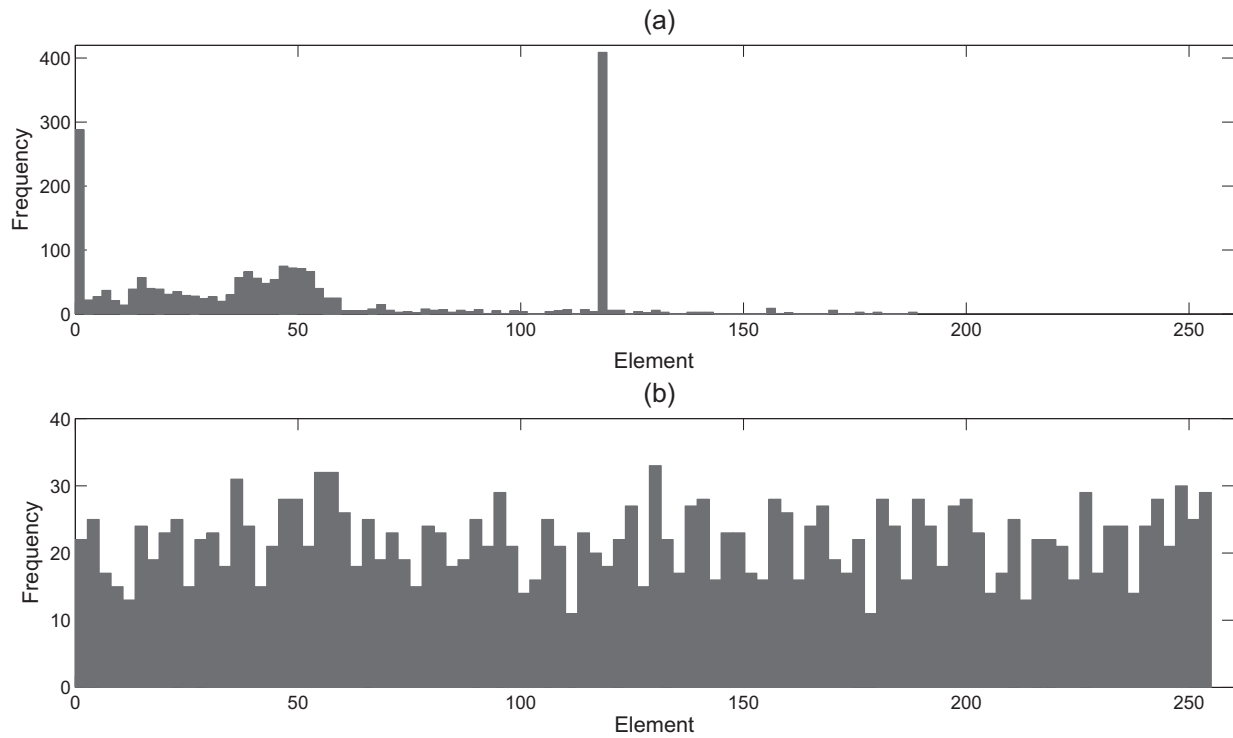


Fig. 10. Histogram analysis: (a) plaintext template and (b) encrypted template.

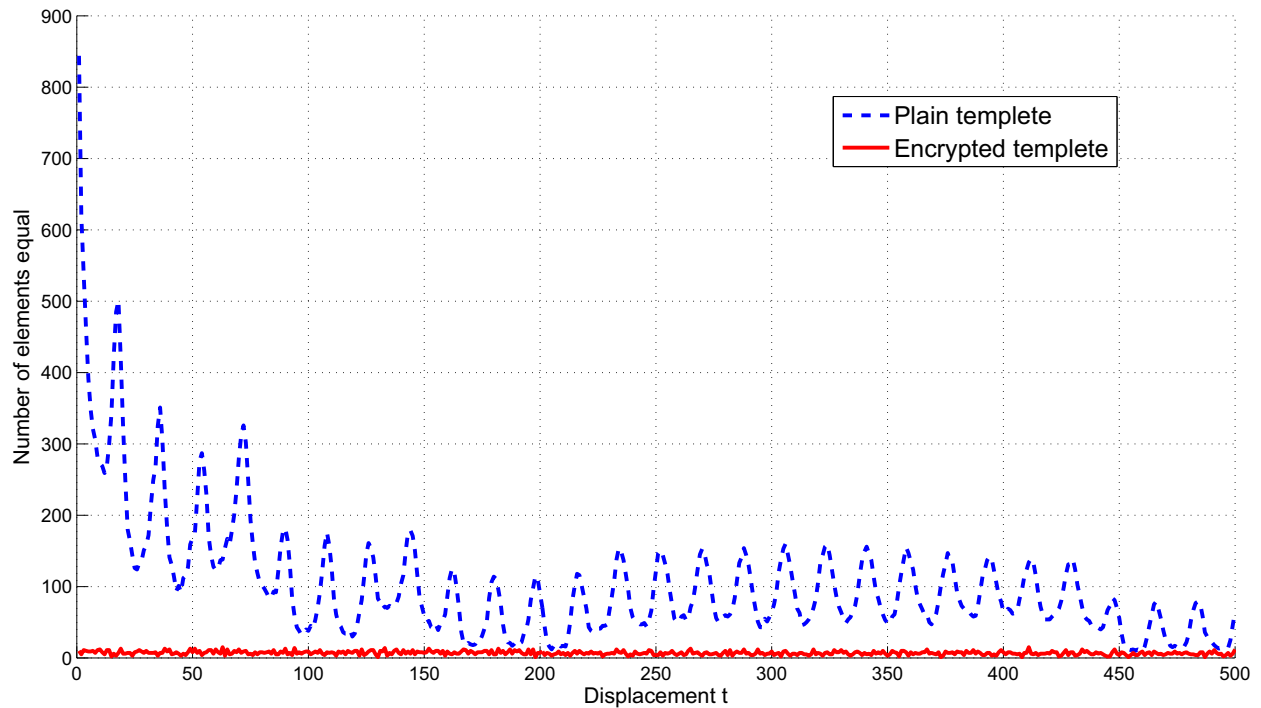


Fig. 11. Autocorrelation analysis of plaintext template (discontinue line), and encrypted template (continue line).

4.7. Information entropy analysis

Information entropy can measure the unpredictability of a random sequence, i.e. how much disorder produces the encryption algorithm at output: greater the unpredictability of the message, greater is the entropy. Other wise, the encryption process is not random and the system can be subject to an successful entropy

attack, due there exist a certain degree of predictability of the encryption method. The entropy $H(T)$ of a template T can be calculated as follows

$$H(T) = \sum_{i=0}^{2^N-1} p(T_i) \log_2 \left(\frac{1}{p(T_i)} \right), \quad (16)$$

where N is the number of bits of each element of the template T , 2^N is all possible symbols in template, $p(T_i)$ represents a probability of the element T_i in the template, and the entropy is expressed in bits. If there is a template T encrypted with 2^N possible elements, the entropy should be $H(T) = N$ ideally. The template has 256 different symbols and its maximum entropy is $H = 8$, this means, all elements appears with the same probability. The entropy of the plain template is $H = 5.47$, whereas the entropy of the encrypted template is $H = 7.91$. Therefore, the encryption template is highly pseudorandom.

4.8. Randomness analysis

The randomness analysis is based in Monobit test, Poker test, Run test, and Serial test. They are used to test if the encryption is a good source of randomness (Menezes, Van Oorschot, & Aanstone, 1996). The results for these test are based on the Chi-Square distribution table with a significance level of $\alpha = 0.05$, and a variable degree of freedom. In addition, each test is based in one hundred cryptograms using the same plain template but one hundred secret keys selected randomly.

The cryptogram should have the same number of zeros and ones ideally (at bit level). Monobit tests how many zeros and ones are in the cryptogram. The Monobit is calculated with the following expression

$$Mt = (\text{zeros} - \text{ones})^2 / (2073 * 8), \quad (17)$$

where Mt is the Monobit test of the encrypted template, zeros is the number of 0's in cryptogram, and ones is the number of 1's in cryptogram. Considering a Chi-Square distribution of one degree of freedom, the critical range for Mt should be less than 3.814; In Fig. 12(a), one hundred Monobit test are shown for each cryptogram and 96% pass the test. The Poker tests if a sequence of m length and all its combinations are reproduced by the same probability, this is determined as follows

$$Pt = \left(\frac{2^m}{k} \right) \left(\sum_{i=1}^{i=c} \text{sum}(i) \right) - k, \quad (18)$$

where Pt is the Poker test, $\text{sum}(i)$ is the occurrence of the sequence i in the cryptogram, $m = 3$ is the sequence length, $c = 2^3$ is all possible combinations of $m \in (000, 001, \dots, 111)$, and $k = (2073 * 8) / 3$ is the non-overlapping trigram in the cryptogram. According to the table for the Chi-Square with $2^m - 1 = 7$ degrees of freedom, the threshold value is 14.07. In Fig. 12(b), one hundred Poker test are shown for each cryptogram and 97% pass the test. In Run test, a run of j ones 0111...10 (the number of contiguous 1's) is known as a *block* and a run of j zeros 1000...01 is called a *gap*; the Run tests if the number of binary runs or ones appear with the same probability in the cryptogram. The following expression is used to calculate the Run test for $j = 1, 2, 3$ length of ones and zeros

$$Rt = \sum_{j=1}^3 \left[\frac{(B_j - E_j)^2}{E_j} \right] + \left[\frac{(G_j - E_j)^2}{E_j} \right], \quad (19)$$

where Rt is the Run test, $E_j = (N - j + 3) / 2^{(j+2)}$ is the expected number of runs of length $j = 1, 2, 3$ in a $N = 8$ -bit sequence, B_j is the number of *blocks* of length j in the cryptogram, and G_j is the number of *gaps* of length j in the cryptogram. With $(2 * j - 2) = 4$ degrees of freedom, the Chi-Square distribution table has the limit of 9.488. In Fig. 12(c), one hundred Run test are shown for each cryptogram and 97% pass the test. The Serial tests if one bit is dependent from its predecessor, this test is given by

$$St = \left(\frac{4}{L} \right) \left((N_{00})^2 + (N_{01})^2 + (N_{10})^2 + (N_{11})^2 \right) - \left(\frac{2}{L} \right) \left((N_0)^2 + (N_1)^2 \right) + 1, \quad (20)$$

where St is the Serial test, $L = (2073 * 8) - 1$ is the length, N_0 and N_1 are the numbers of zeros and ones in the cryptogram, and $N_{00,01,10,11}$ is the relevant numbers of sub-sequences with overlap. The

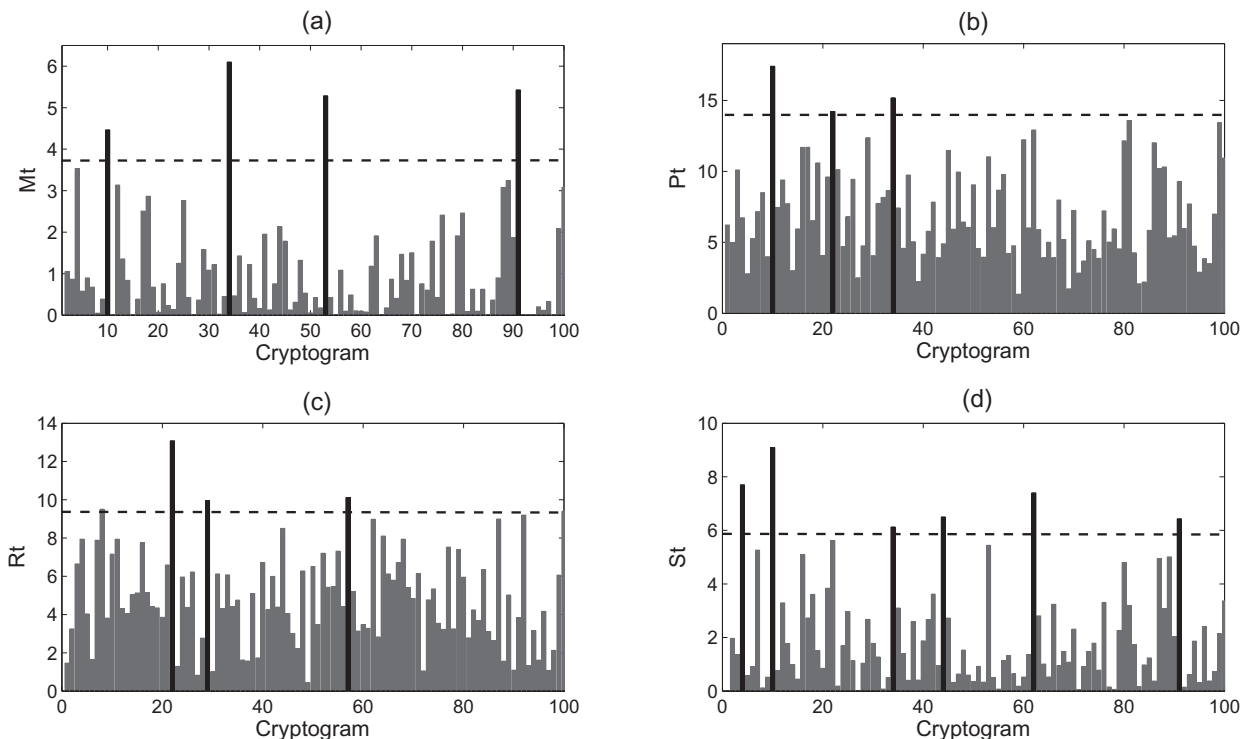


Fig. 12. Randomness analysis of one hundred cryptograms: (a) Monobit test, (b) Poker test, (c) Run test, and (d) Serial test.

Table 4

Comparison of the proposed fingerprint protection scheme with similar approaches in literature.

	Scheme				
	Proposed	Supriya and Manjunatha (2014)	Mihailescu (2014)	Ghouzali and Abdul (2013)	Li et al. (2012)
Classification technique	Feature transform. Chaotic encryption based on logistic map	Feature transform. Chaos and cancellable biometrics based on logistic map	Feature transform. Chaos and Hash functions based on Rossler map and logistic map	Feature transform. Chaotic encryption based on logistic map	Feature transform. Chaos and cancellable biometrics based on FFNDF system
Biometric template	Fingerprint	–	Fingerprint	Face feature	Palmprint
Matching domain	Plain domain	Transformed domain	Transformed domain	Transformed domain	Transformed domain
Revocability	Not	Yes	–	Yes	Yes
Diversity	Yes	Yes	–	Yes	Yes
Security	Yes	–	–	Yes	Yes
FAR and FRR affected	Not	–	–	–	Not
<i>Security analysis</i>					
Secret key space	2^{128} all strong	2^{319}	–	–	2^{210}
Secret key sensitivity	Yes	Yes	–	–	Yes
Chaotic data optimized	Yes	Not	–	–	–
Plain template sensitivity	Yes	–	–	–	–
Floating frequency	Yes	–	–	–	–
Histogram uniform	Yes	–	–	–	–
Autocorrelation	Yes	–	–	–	–
Entropy	Yes	Yes	–	–	–
Randomness	Yes	–	–	–	–
Chosen/known plain template attack	Yes	–	–	–	–
<i>Implementation</i>					
Enrollment time (s)	2.2	–	–	–	–
Authentication time (s)	3.1	–	–	–	0.2
Cost (USD)	250	–	–	–	–

Chi-Square distribution with two degrees of freedom is given by 5.991, and Fig. 12(d) shows the S_t results over one hundred cryptograms with 94% successful test.

All the previous tests verify the high randomness of the encryption process. In addition, the results show that more than 95% of all possible secret key can be considered highly secure and the another 5% can be considered half-strong, but not weak keys due it can generates good pseudorandom properties.

4.9. Implementation and performance analysis

The implementation is based in a 32 bit COLDFIRE microcontroller M52259 Freescale. The encryption algorithm is programmed in code C with CodeWarrior software. The microcontroller uses a frequency system of 48 MHz, RS-232 (for fingerprint module) and I2C (for matrix keypad and LCD modules) communication ports, Flash memory, and output ports for control access activation. The algorithm implementation requires 116 KB (22%) of 512 KB Flash memory. The 400 KB are used to store encrypted template from authorized users. Therefore, the proposed authentication system can register until 370 users with just the microcontroller Flash memory.

The proposed authentication system has the fingerprint recognition accuracy of *false acceptance rate* FAR 10^6 (one in a million) and *false rejection rate* FRR 10^2 (one in one hundred) based on fingerprint module Futronic FS83 specifications. The registration time of a new user in the embedded system requires just 2.2 seconds including read user's fingerprint, encryption in microcontroller, and storage in Flash memory. The authentication time is 3.1 seconds for read the user's fingerprint, read the user's fingerprint from Flash memory, decryption process, and the matching process in fingerprint module. The implementation cost are lower with \$250 usd. Therefore, it can be used in real time applications as embedded expert systems.

4.10. Comparison with similar schemes in literature

The proposed fingerprint template protection is based on feature transform, i.e. all plain template is transformed into uncorrelated template. In contrast with schemes related with feature transform in literature, our scheme needs decipher the encrypted template in matching process. Therefore, the matching process is performed in plain domain against the literature's schemes, which perform the matching process in transform domain to keep the biometric template revocable if it is compromised.

In our scheme, the biometric trait could be lost if the encryption algorithm is broken or the secret key is known by an adversary. Nevertheless, we can achieve an exhaustive security analysis to achieve diversity, security and performance. The security properties of the proposed encryption algorithm make each fingerprint protected template unique without affect the FAR and FRR rates. Table 4 shows a comparison with several biometric protection schemes in literature.

5. Conclusion

In this work, we proposed a fingerprint template protection based on chaotic encryption. In addition, the proposed scheme was implemented in an embedded authentication system based on a 32 bit microcontroller. In contrast with another protection template schemes in literature, we verify the security of our approach by a complete security analysis at statistical level, which verify and justify the high capabilities of our scheme in security aspects and its implementation in real applications embedded systems.

The principal disadvantage of our scheme is the irrevocability of the biometric trait, since we need perform the matching process at plain domain level. Nevertheless, our approach can be used in

sensitive applications due it presents excellent encryption properties, which were tested and verified in this work.

The proposed scheme has high potential in several applications in both simple embedded system or embedded expert systems such as control access in offices, banks, factories, hospitals, universities, homes, e-commerce, cellphones, personal systems and others.

Future work needs to be considered to increase the security and performance capabilities in an embedded expert system. For example add to the system the capability to use two or more biometrics, i.e. voice recognition with fingerprint template protection by using our scheme (this makes the embedded system “intelligent” to determine if the user is not an impostor). In addition, the propose fingerprint protection scheme can be used in high computational power systems such as FPGA, where a digital video camera can be implemented to give to the system the ability of “seeing” the user face in real-time and “close the system” if he/she is not recognized in this stage. Another real-time monitoring schemes could be implemented such as thermal sensing, heart rate sensing, pulse rate sensing, etc. which can improve the security of the embedded expert system to determine if an authorized live person is using the system, and not an intruder or a malicious software.

Acknowledgement

This work was supported by the CONACYT, Mexico under Research Grant 166654.

References

- Abbasy, M. R., Manaf, A. A., & Shahidan, M. A. (2011). Data hiding method based on DNA basic characteristics. *Communications in Computer and Information Science*, 194, 53–62.
- Abhyankar, A., Vijayar, A., Kumar, S., & Schuckers, S. (2009). Encryption of biometric templates using one time biometric transform algorithm. *CiiT International Journal of Biometrics and Bioinformatics*, 1–8.
- Adleman, L. M. (1994). Molecular computation of solution to combinatorial problems. *Science*, 266(5187), 1021–1024.
- Admek, M., Matsek, M., & Neumann, P. (2015). Security of biometric systems. *Procedia Engineering*, 100, 169–176.
- Al-Assam, H., Sellahewa, H., & Jassim, S. (2009). A Lightweight approach for biometric template protection. In *Proceedings of SPIE mobile multimedia/image processing, security, and applications, Florida, USA* (p. 7351).
- Alvarez, G., & Li, S. (2006). Some basic cryptographic requirements for chaos-based cryptosystems. *International Journal of Bifurcation and Chaos*, 16(8), 2129–2151.
- Alvarez, G., & Li, S. (2009). Cryptanalyzing a nonlinear chaotic algorithm (NCA) for image encryption. *Communications in Nonlinear Science and Numerical Simulation*, 14(11), 3743–3749.
- Arakala, A., Jeffers, J., & Horadam, K. J. (2007). Fuzzy extractors for minutiae-based fingerprint authentication. In *Proceedings of the 2nd international conference on biometrics*, Seoul, South Korea (p. 760769).
- Arroyo, D., Alvarez, G., & Fernandez, V. (2008). On the inadequacy of the logistic map for cryptographic applications. *X Reunin Española sobre Criptología y Seguridad de la Informacin*, 77–82.
- Balaji, N., & Nagaraj, N. (2008). Cryptanalysis of a chaotic image encryption algorithm. In *National conference on nonlinear systems and dynamics* (Vol. 1, pp.1–14).
- Bhatnagar, G. (2012). Chaos-based security solution for fingerprint data during communication and transmission. *IEEE Transactions on Instrumentation and Measurement*, 61(4), 876–887.
- Bhatnagar, G., & Jonathan Wu, Q. M. (2012). Chaos-based security solution for fingerprint data during communication and transmission. *IEEE Transactions on Instrumentation and Measurement*, 61(4), 876–887.
- Bhatnagar, G., & Jonathan Wu, Q. M. (2015). A novel chaos-based secure transmission of biometric data. *Neurocomputing*, 147, 444–455.
- Bordoa, M., & Tornea, O. (2010). DNA secret writing techniques. In *8th International Conference on communications* (pp. 451–456).
- Cavoukian, A., & Stoianov, A. (2007). Biometric encryption: A positive-sum technology that achieves strong authentication, security and privacy. *Information and Privacy Commissioner of Ontario*, 48.
- Chang, E. C., & Roy, S. (2007). Robust extraction of secret bits from minutiae. *Proceedings of 2nd international conference on biometrics*, Seoul, South Korea (p. 750759).
- Chen, C. -H., & Dai, J. -H. (2005). An embedded fingerprint authentication system with reduced hardware resources requirement. In *Proceedings of the ninth international symposium on consumer electronics* (pp. 145–150).
- Cokal, C., & Solak, E. (2009). Cryptanalysis of a chaos-based image encryption algorithm. *Physics Letters A*, 373(15), 1357–1360.
- Cristian-Iulian, R., & Vasile-Gabriel, I. (2014). Aspects regarding chaotic maps hardware implementations. *Revue Roumaine Des Sciences Techniques*, 52(2), 219–227.
- Cui, D. (2010). A novel fingerprint encryption algorithm based on chaotic system and fractional fourier transform. In *IEEE international conference on machine vision and human-machine interface* (pp. 168–171).
- Ferguson, N., Schroepel, R., & Whiting, D. (2001). A simple algebraic representation of Rijndael. *Lecture Notes in Computer Science*, 2259, 103–111.
- Fons, M., Fons, F., & Cantó, E. (2007). Embedded security: New trends in personal recognition systems. In *Research in microelectronics and electronics conference* (pp. 89–92).
- Ghouzali, S., & Abdul, W. (2013). Private chaotic biometric template protection algorithm. In *Proceedings of the IEEE second international conference on image information processing* (pp. 655–659).
- Guozhen, X., Mingxin, L., Lei, Q., & Xuejia, L. (2006). New field of cryptography: DNA cryptography. *Chinese Science Bulletin*, 51(12), 1413–1420.
- Han, F., Hu, J., Yu, X., & Wang, Y. (2007). Fingerprint images encryption via multi-scroll chaotic attractors. *Applied Mathematics and Computation*, 185, 931939.
- Hsiao, J.-I., & Lee, J. (2015). Fingerprint image cryptography based on multiple chaotic system. *Signal Processing*, 113, 169–181.
- Imamverdiyev, Y., Beng Jin Teoh, A., & Kim, J. (2013). Biometric cryptosystem based on discretized fingerprint texture descriptors. *Expert Systems with Applications*, 40, 18881901.
- Inuma, M., & Otsuka, A. (2013). Relations among security metrics for template protection algorithms. In *IEEE sixth international conference on biometrics: Theory, applications and systems*, Arlington, VA (pp. 1–8).
- Inzunza-González, E., & Cruz-Hernández, C. (2013). Double hyperchaotic encryption for security in biometric systems. *Nonlinear Dynamics and Systems Theory*, 13(1), 55–68.
- Jain, A. K., Nandakumar, K., & Nagar, A. (2008). Biometric template security. In *EURASIP journal on advances in signal processing* (pp. 17). springer.
- Jain, A. K., Nandakumar, K., & Nagar, A. (2012). Fingerprint template protection: From theory to practice. In *Security and privacy in biometrics* (pp. 187–214). London: Springer.
- Jain, A. K., & Uludag, U. (2003). Hiding biometric data. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 25(11), 1494–1498.
- James, D., & Philip, M. (2012). A novel face template protection scheme based on chaos and visual cryptography. *International Journal of Applied Information Systems*, 2(5), 31–35.
- Jin, Z., Beng Jin Teoh, A., Song Ong, T., & Tee, C. (2012). Fingerprint template protection with minutiae-based bit-string for security and privacy preserving. *Expert Systems with Applications*, 39, 61576167.
- Kaur, M., Sofat, S., & Saraswat, S. (2010). Template and database security in biometrics systems: A challenging task. *International Journal of Computer Applications*, 4(5), 1–5.
- Kevenaar, T., Korte, U., Merkle, J., Niesing, M., Ihmor, H., Busch, C., & Zhou, X. (2010). A reference framework for the privacy assessment of keyless biometric template protection systems. In *Proceedings of BIOSIG* (pp. 45–56).
- Khan, M. K., & Zhang, J. (2005). Securing biometric templates for reliable identity authentication. In *International conference on intelligent computing*, Hefei, China (pp. 23–26).
- Khan, M. K., Zhang, J., & Tian, L. (2004). Protecting biometric data for personal identification. *Sinobiometrics*, 629638.
- Khan, M. K., Zhang, J., & Tian, L. (2007). Chaotic secure content-based hidden transmission of biometric templates. *Chaos, Solitons and Fractals*, 32, 1749–1759.
- Li, C., Chen, G., & Halang, W. A. (2009). Cryptanalysis of an image encryption scheme based on a compound chaotic sequence. *Image and Vision Computing*, 27(8), 1035–1039.
- Liu, R. (2012). Chaos-based fingerprint images encryption using symmetric cryptography. In *9th International conference on fuzzy systems and knowledge discovery* (pp. 2153–2156).
- Li, H., Wang, L., Zhang, R., & Wu, L. (2012). New enrollment scheme for biometric template using hash chaos-based cryptography. *Journal of Software*, 7(8), 1827–1834.
- Li, H.-J., & Zhang, J.-S. (2010). A novel chaotic stream cipher and its application to palmprint template protection. *Chinese Physics B*, 19(4), 1–10.
- May, R. M. (1976). Simple mathematical models with very complicated dynamics. *Nature*, 261(5560), 459–467.
- Menezes, A. J., Van Oorschot, P. C., & Aanstone, V. S. (1996). In *Handbook of applied cryptography* (pp. 810). CRC Press.
- Mihailescu, M. I. (2014). New enrollment scheme for biometric template using hash chaos-based cryptography. *Procedia Engineering*, 69, 14591468.
- Militello, C., Conti, V., Vitabile, S., & Sorbello, F. (2008). A novel embedded fingerprints authentication system based on singularity points. In *International conference on complex, intelligent and software intensive systems* (pp. 72–78).
- Moujahdi, C., Bebis, G., Ghouzali, S., & Rziza, M. (2014). Fingerprint shell: Secure representation of fingerprint template. *Pattern Recognition Letters*, 45, 189196.
- Murillo-Escobar, M. A., Cruz-Hernández, C., Abundiz-Pérez, F., López-Gutiérrez, R. M., & Acosta Del Campo, O. (2015). A RGB image encryption algorithm based on total plain image characteristics and chaos. *Signal Processing*, 109, 119–131.
- Murillo-Escobar, M. A., Abundiz-Pérez, F., Cruz-Hernández, C., & López-Gutiérrez, R. M. (2014). A novel symmetric text encryption algorithm based on logistic map. In *Proceedings of the international conference on communications, signal processing and computers* (pp. 49–53).

- Nandakumar, K., Nagar, A., & Jain, A. K. (2007). Hardening fingerprint fuzzy vault using password. *Lecture Notes in Computer Science*, 4642, 927–937.
- Nayak, D. R. (2008). A novel architecture for embedded biometric authentication system. *European Symposium on Computer Modeling and Simulation*, 567–572.
- Nie, T., Li, Y., Zhou, L., & Toyonaga, M. (2013). A multilevel fingerprinting method for FPGA IP protection. *IEEE International Symposium on Circuits and Systems*, 1789–1792.
- Munaga, V. N. K. P., & Santhosh Kumar, C. (2014). Fingerprint template protection using multiline neighboring relation. *Expert Systems with Applications*, 41, 61146122.
- Ratha, N. K., Chikkerur, S., Connell, J. H., & Bolle, R. M. (2007). Generating cancelable fingerprint templates. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 29(4), 561–572.
- Ratha, N. K., Connell, J. H., & Bolle, R. M. (2001). An analysis of minutiae matching strength. *Lecture Notes in Computer Science*, 2091, 223–228.
- Rhouma, R., Solak, E., & Belghith, S. (2010). Cryptanalysis of a new substitution-diffusion based image cipher. *Communications in Nonlinear Science and Numerical Simulation*, 15(7), 1887–1892.
- Shannon, C. E. (1949). Communication theory of secrecy systems. *The Bell System Technical Journal*, 28, 656–715.
- Solak, E., Cokal, C., Yildiz, O. T., & Biyikiglu, T. (2010). Cryptanalysis of Fridrich's chaotic image encryption. *International Journal of Bifurcation and Chaos*, 20(5), 1405–1413.
- Soutar, C., Roberge, D., Stoianov, A., Gilroy, R., & Vijaya Kumar, B. V. K. (1999). *Biometric encryption. ICSA guide to cryptography*. McGraw-Hill.
- Supriya, V. G., & Manjunatha, S. R. (2014). Chaos based cancellable biometric template protection scheme—a proposal. *International Journal of Engineering Science Invention*, 3(11), 2319–6726.
- Sutcu, Y., Li, Q., & Memon, N. (2007). Protecting biometric templates with sketch: Theory and practice. *IEEE Transactions on Information Forensics and Security*, 2(3), 503512.
- Taylor, C., Risca, V., & Bancroft, C. (1999). Hiding messages in DNA microdots. *Nature*, 399(6736), 533–534.
- Teoh, A. B. J., Goh, A., & Ngo, D. C. L. (2006). Random multispace quantization as an analytic mechanism for BioHashing of biometric and random identity inputs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 28(12), 1892–1901.
- Teoh, A. B. J., & Kim, J. (2007). Secure biometric template protection in fuzzy commitment scheme. *IEICE Electronics Express*, 4(23), 724–730.
- Uludag, U., Pankanti, S., Prabhakar, S., & Jain, A. K. (2004). Biometric cryptosystems: Issues and challenges. *Proceedings of the IEEE*, 92(6), 948–960.
- Wang, Y., Liao, X., Xiang, T., Wong, K.-W., & Yang, D. (2007). Cryptanalysis and improvement on a block cryptosystem based on iteration a chaotic map. *Physics Letters A*, 363(4), 277–281.
- Wang, Y., Li, D., Isshiki, T., & Kunieda, H. (2004). A novel SOC architecture embedded bit serial FPGA. *IEEE Conference on Circuits and Systems*, 1, 133–136.
- Xiaomin, W., TaiHua, X., & Wenfang, Z. (2011). Chaos-based biometrics template protection and secure authentication. In Jucheng Yang, Dr. (Ed.), *State of the art in biometrics*. ISBN: 978-953-307-489-4.
- XueJia, L., MingXin, L., Lei, Q., JunSong, H., & XiWen, F. (2010). Asymmetric encryption and signature method with DNA technology. *Science China Information Sciences*, 53(3), 506–514.
- Zhao, S., Li, H., & Yan, X. (2008). A secure and efficient fingerprint images encryption scheme. In *The 9th international conference for young computer scientists* (pp. 2803–2808).
- Zhou, S., Zhang, Q., Wei, X., & Zhou, C. (2010). A summarization of image encryption. *IETE Technical Review*, 27(6), 503–510.